

Mobile measurements



NextAIRE

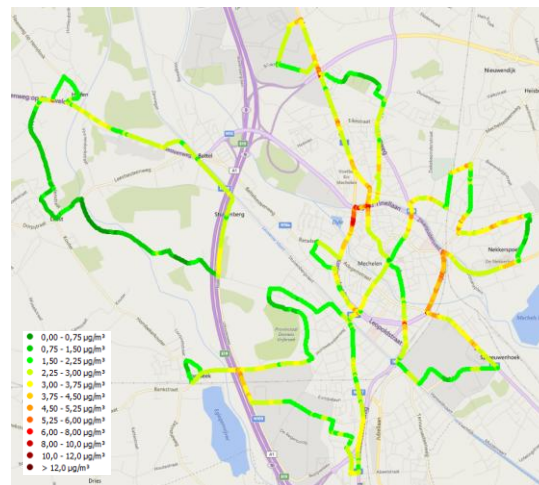
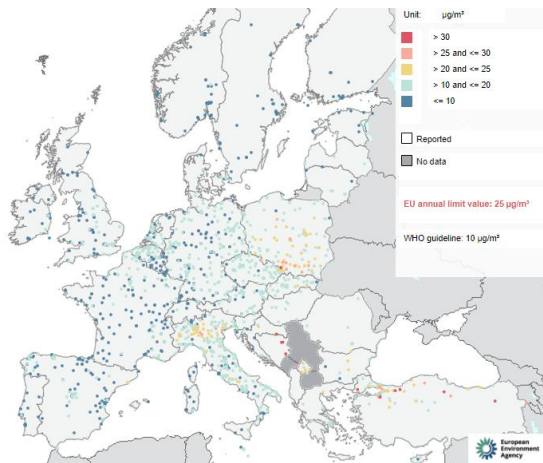
Martine Van Poppel
VITO

**Sarajevo Summer School
7-8 September 2026**

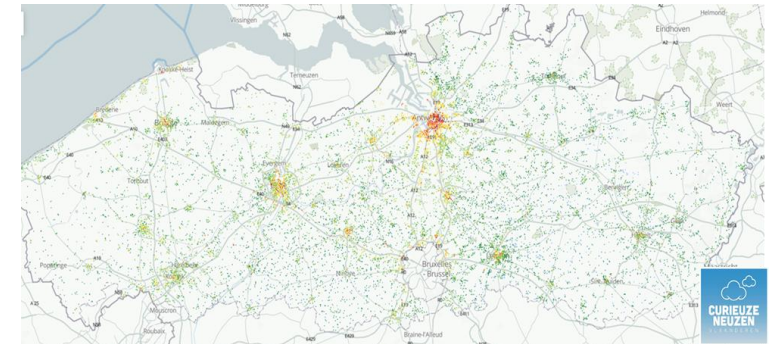


Why Urban AQ mapping

- Fixed AQM stations
 - Accurate
 - Long time series, time trends
 - BUT limited in spatial resolution
- Air Quality is **not** uniform over a city, a region
 - Insufficient info about exposure and health effects
 - Source apportionment
- Health studies/policy/... require high spatial (and temporal) resolution data
- Scientific approach and data quality
- (How to involve citizens?)

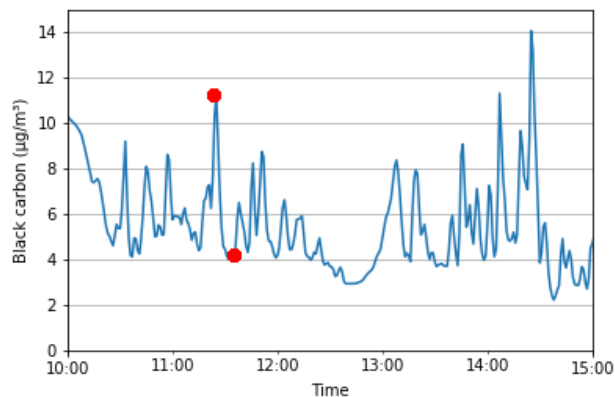


RI-URBANS (101036245)

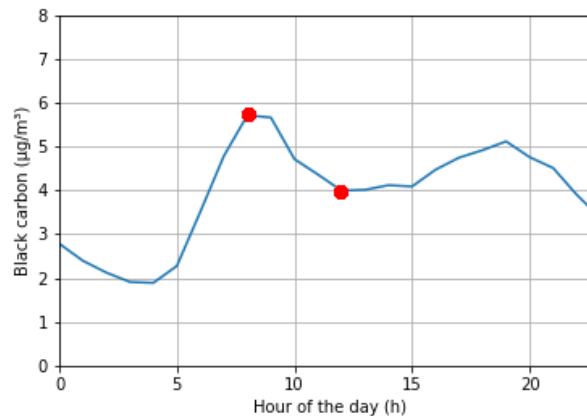


Mobile measurements

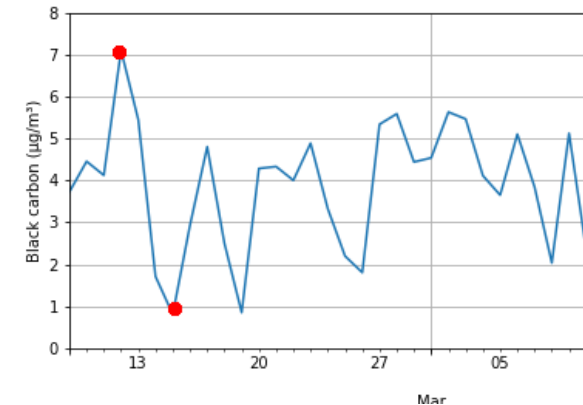
- Urban air quality => large temporal and spatial variability
- Spatio-temporal data => limited temporal resolution
- Repetitions are needed



=>Events, e.g. passing car, ...



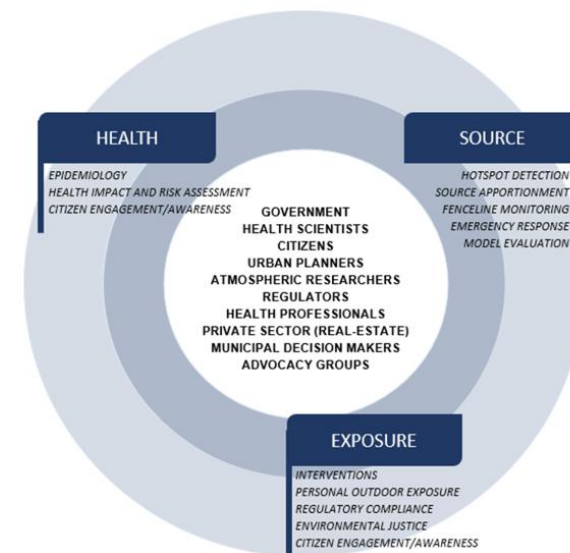
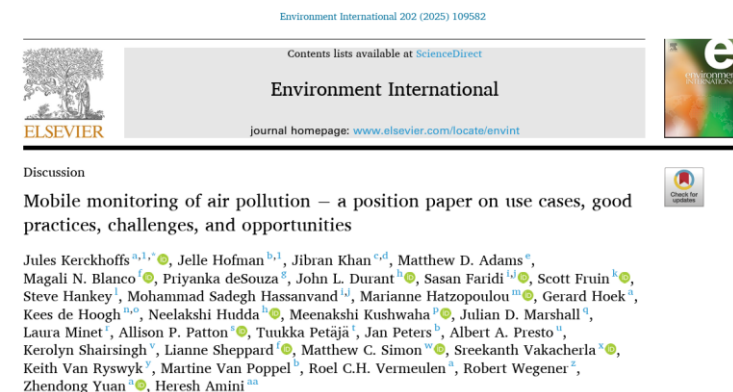
=> Daily pattern (~ sources)



=> Meteorological conditions,
seasonal variation, ...

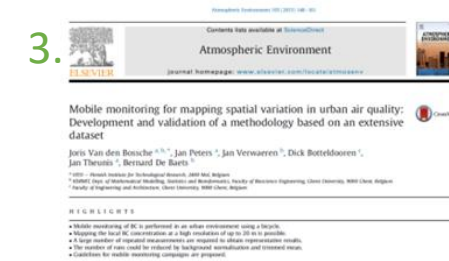
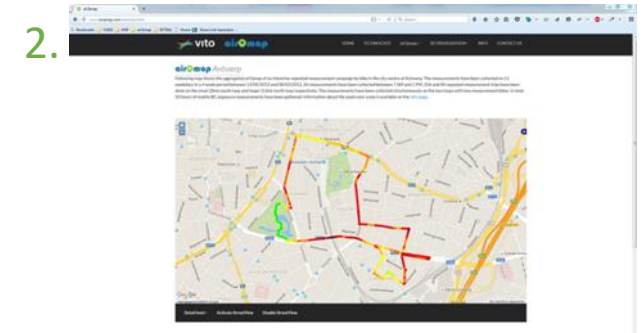
Mobile mapping of air pollution

- Mobile monitoring as efficient tool to complement fixed AQMN by adding spatial resolution
- Experimental design is important and is function of the use case
- Mobile approaches – experimental design:
 - **Dedicated vs opportunistic**
 - Mobile platform: **vehicle vs portable**
 - Instruments - routes
 - **Temporal coverage:** season (sources, dispersion, meteo,...), time of day
- Data processing: data-only vs modelling



Mobile BC monitoring using **airQmap**

- **airQmap**: a tool to collect mobile BC measurements and process them into **street-level BC maps**.
- It contains:
 1. Easy to use **measurement devices** to allow citizens/volunteers to collect mobile BC measurements in a 'cost-effective' way
 2. An automated **data processing** infrastructure and **visualization** to generate the BC map www.airqmap.com
 3. Measurement **methodology** (data collection, data validation and aggregation)
- Has been used in different municipalities and cities



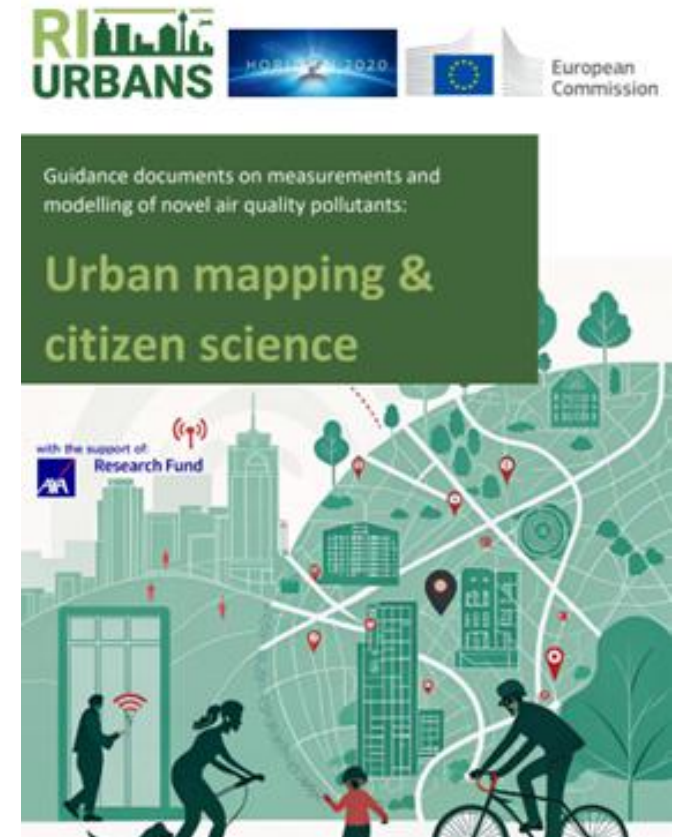
Mobile mapping: extending to other pollutants

- PM mass: $PM_{2.5}$, PM_{10} using sensors
- UFP (PNC) using handheld instrument (Partector)
- Multi wavelength AE: to distinguish between traffic emissions and biomass burning
- New types of instruments (BC): AE51 -> AL30



Urban mapping: mobile measurements or LCS

- Substantial new information can be learned by campaigns of mobile monitoring or low-cost sensor (LCS) fixed site monitoring
 - limited duration (weeks to months)
 - with a proper design
- These methods are useful as an addition to routine monitoring with reference grade instruments at a small number of fixed sites
 - Longer campaigns can result in more precise estimates of air pollution patterns
- Participatory approach is valuable tool to generate a lot of data



Mapping air pollution

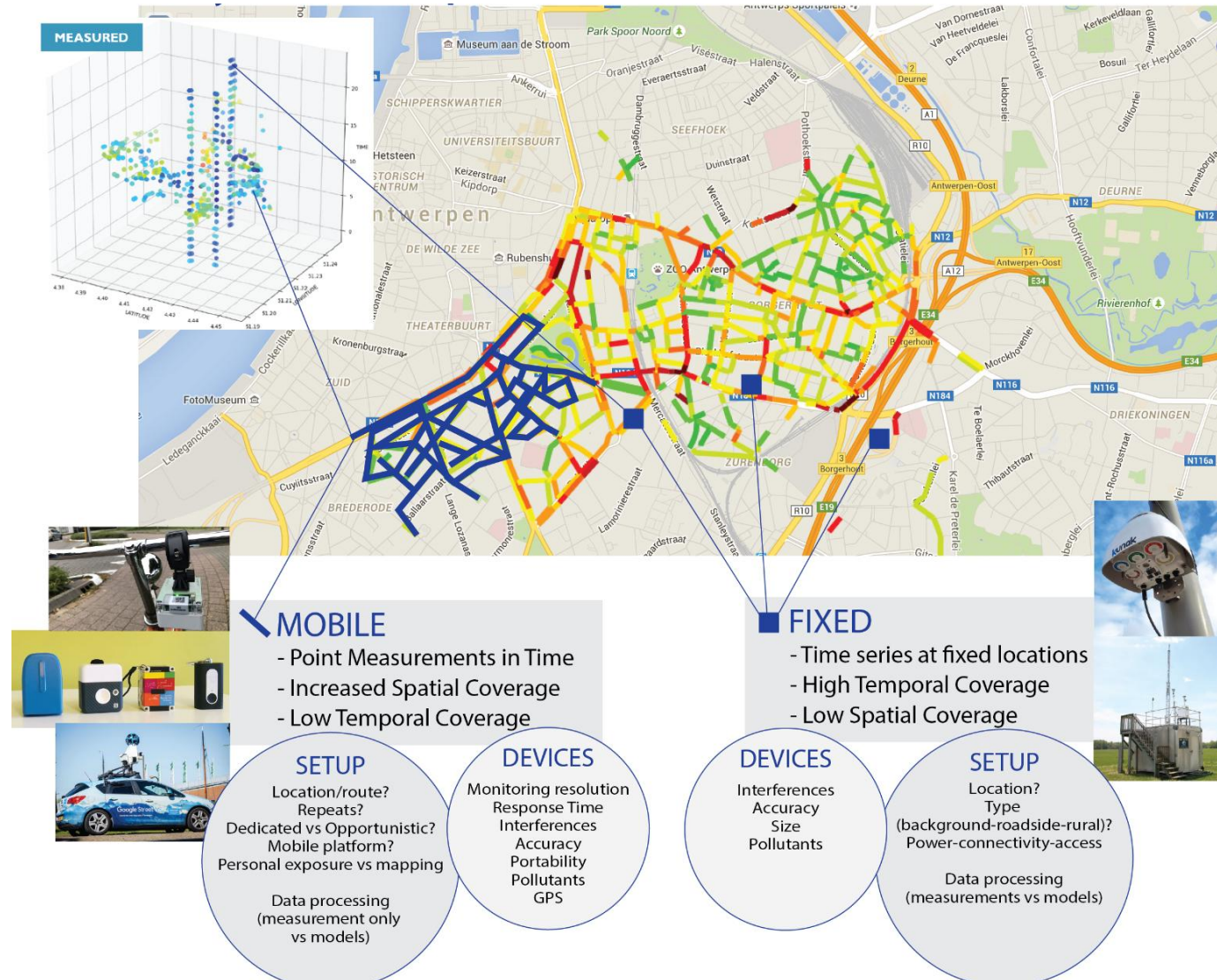
APPROACHES

- Mobile measurements
 - Dedicated measurement approach
 - Opportunistic measurements
- Fixed sensor networks
 - More instruments needed -> lower-cost sensors
 - Time series
 - Data quality of lower-cost sensors

What needs to be defined

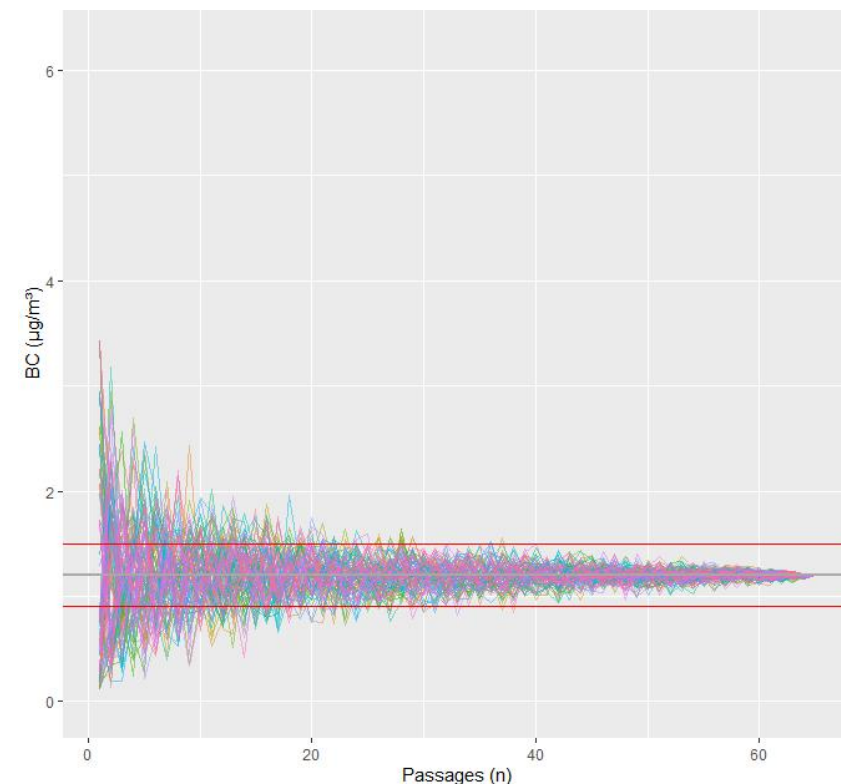
- Monitoring devices
- Monitored pollutants
- Data quality
- Calibration approach
- Monitoring set-up
- Data processing
 - Aggregation
 - (background) correction
 - Calibration factors
 - ...
- Integration of data in AQMN
- Training, feedback

Monitoring strategy and instrumentation (1/2)

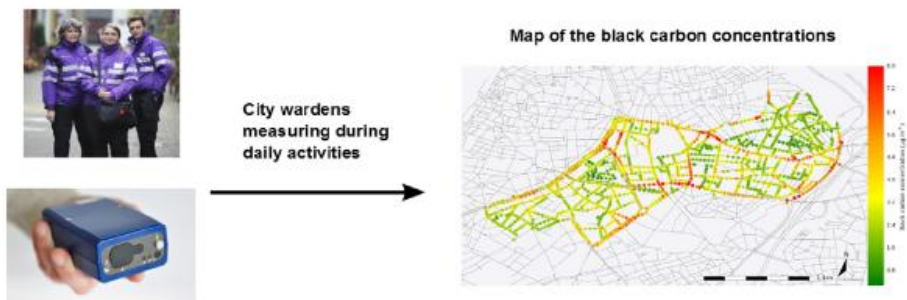


Monitoring strategy and instrumentation (2/2)

- Mobile approach:
 - **Dedicated** vs **opportunistic**
 - **Vehicle** vs **portable**
 - **Different seasons** to capture seasonal variability (sources, dispersion, meteo,...)
 - Minimum **number of repeats!**
 - Period of interest: peak hours, working days, annual average...
- Stationary approach: low-cost sensors
 - Quality control!
- Stationary approach: diffusive tubes
 - Low time resolution



Mobile measurements



airqmap (www.airqmap.com)



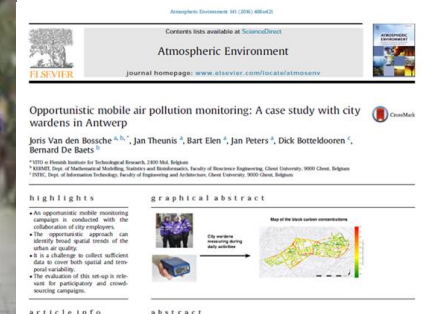
HOPE

RI-URBANS (101036245)



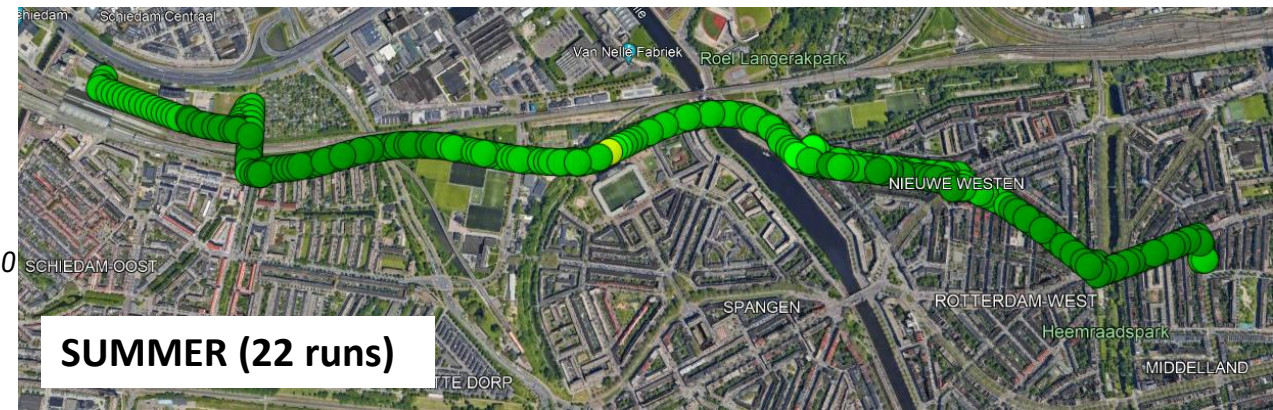
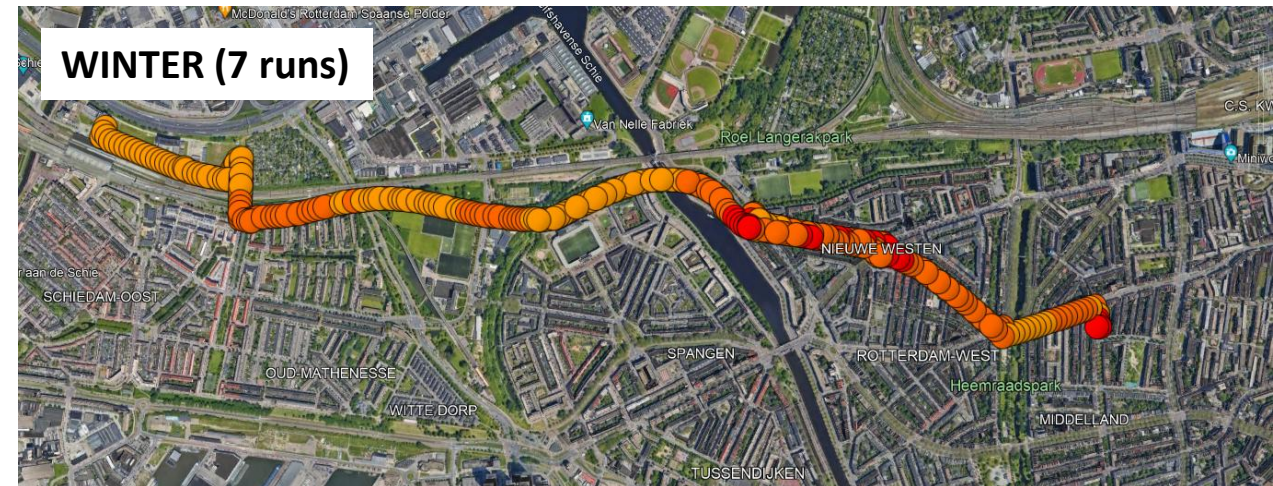
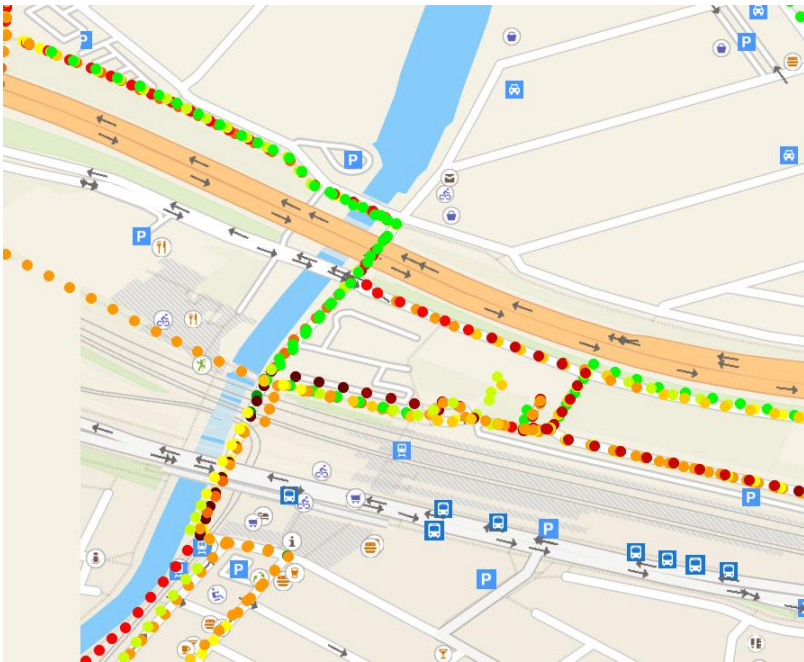
Dedicated versus opportunistic approach

- Dedicated measurement set-up
 - Measurement route is defined (in function of question)
 - At specific time(s) of day
 - Gives a representative picture of these moments in time
 - Easy to compare different locations
 - Additional effort to collect data
- Opportunistic measurements
 - Measurements are performed using a 'platform'
 - Collection of spatio-temporal data
 - Additional effort/attention to data processing
 - Risk of sampling bias



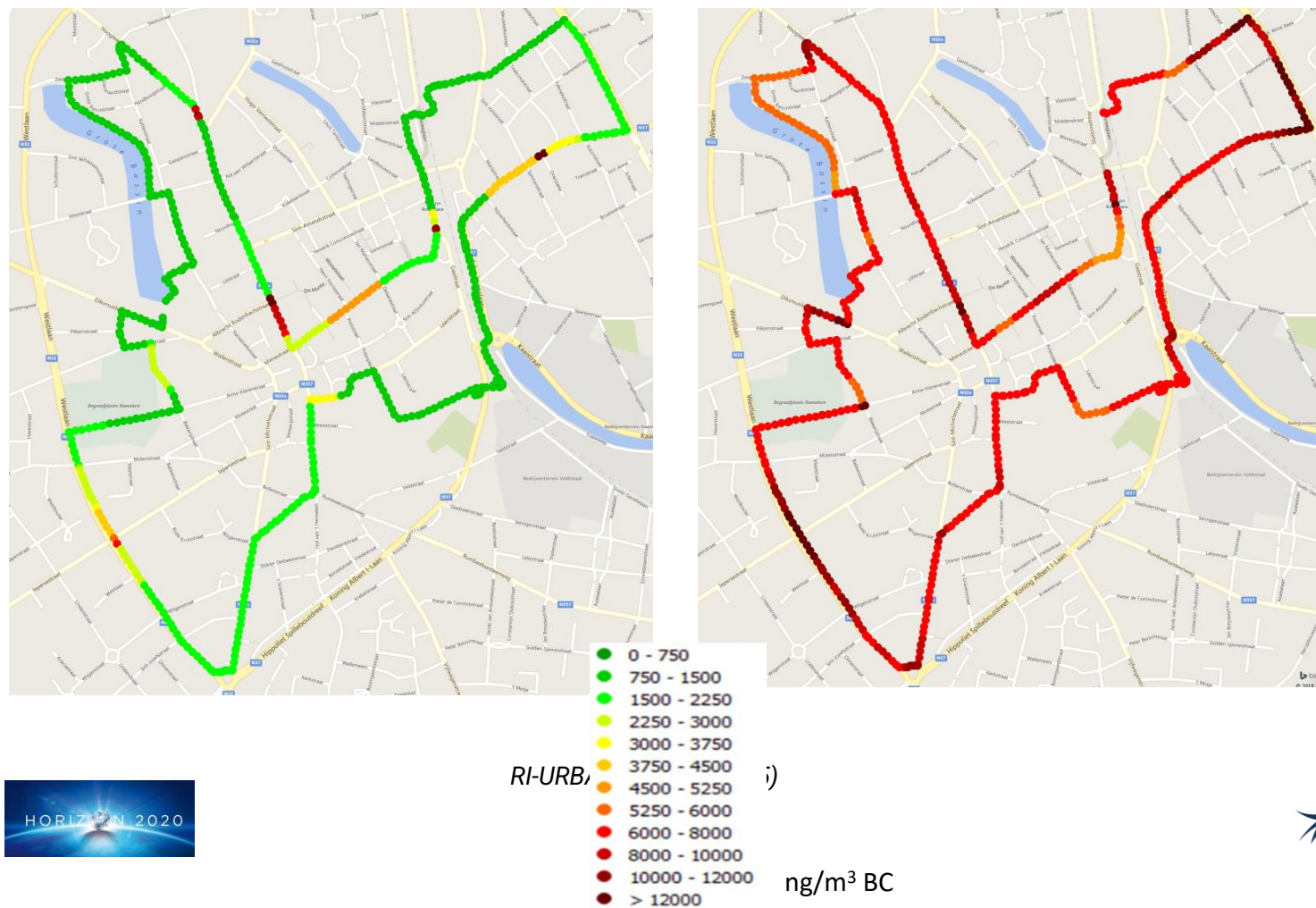
Data processing needs: normalisation

- High day-to-day/seasonal variability → fluctuating background concentration
→ need for normalization!

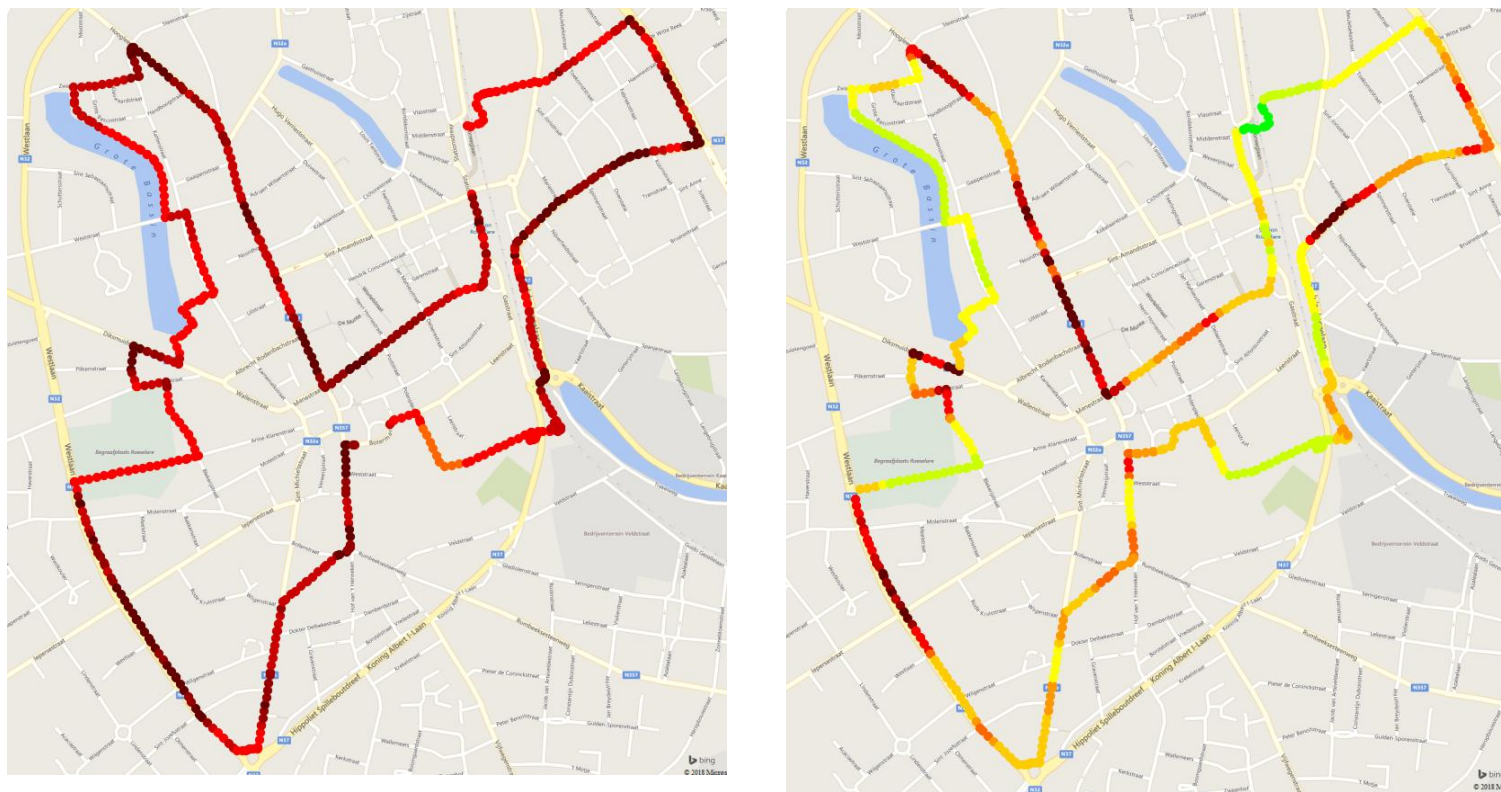


RI-URBANS (10

- 25/10 (morning) and 7/11 (morning) -> day-to-day differences at same time of day



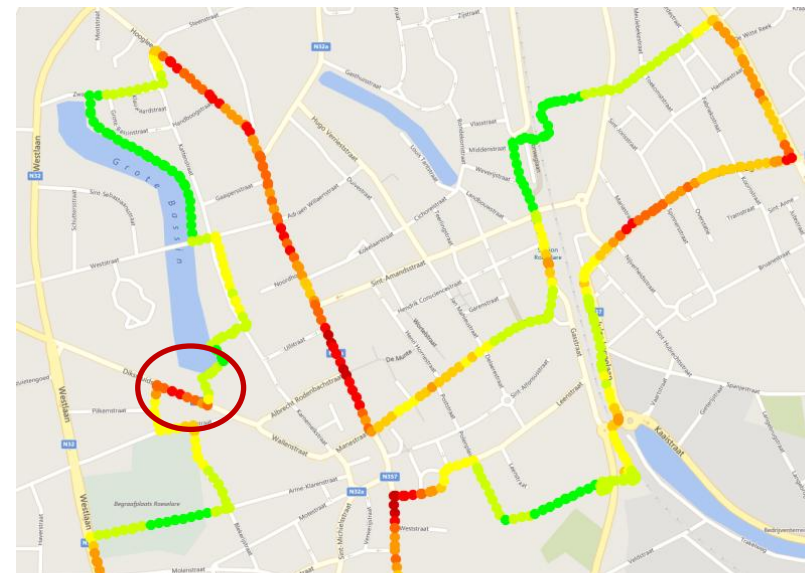
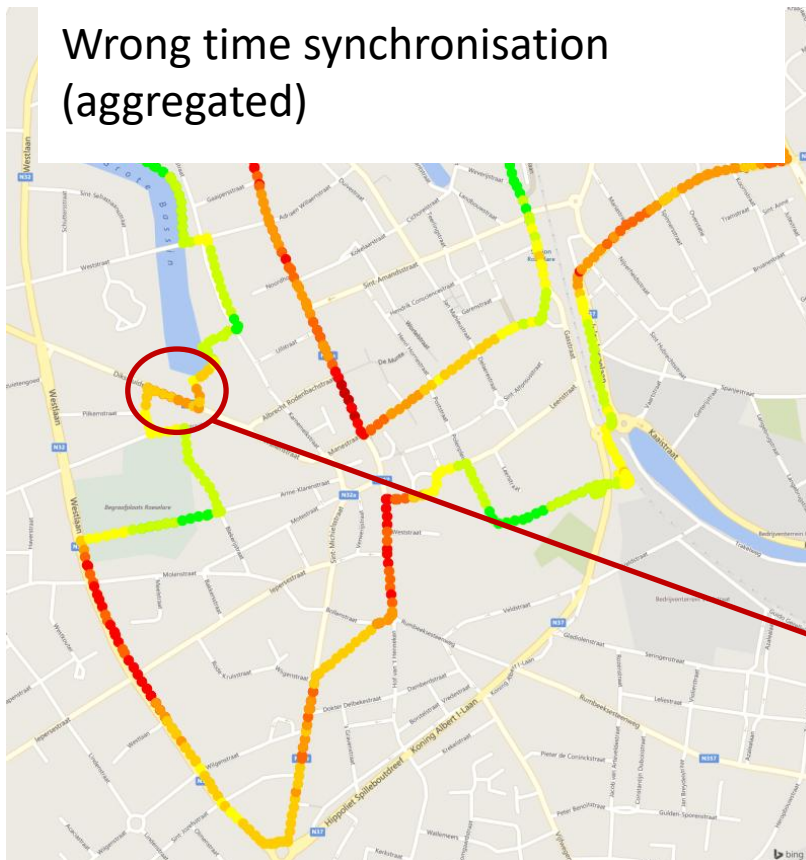
- 8/11 en morning versus afternoon (same day)



RI-URBANS (101036245)

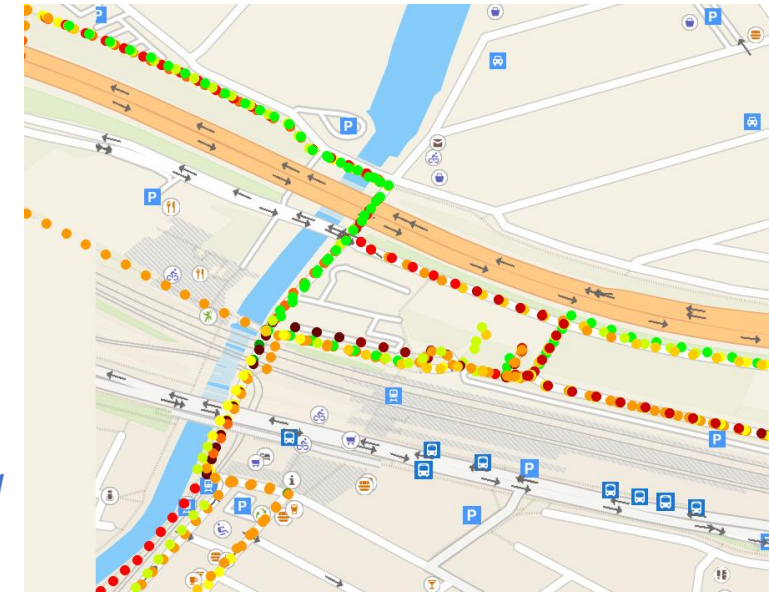
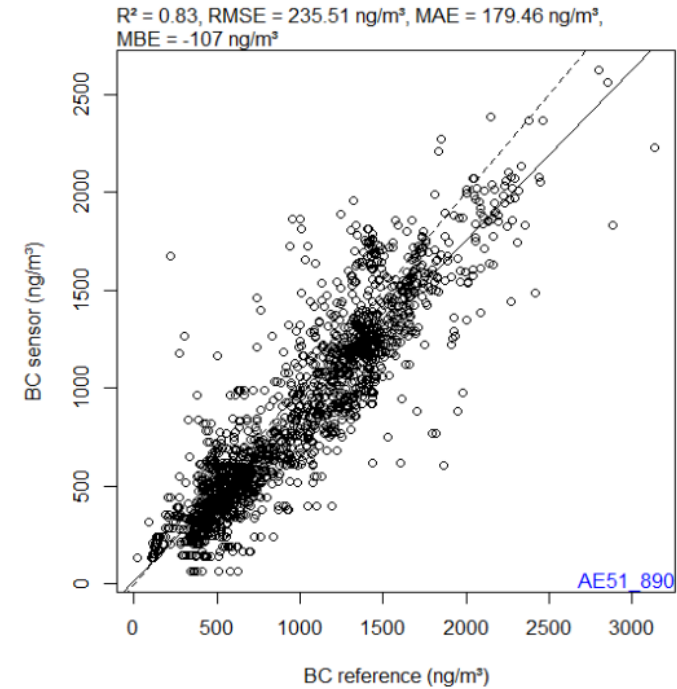
Time synchronization!

Wrong time synchronisation
(aggregated)



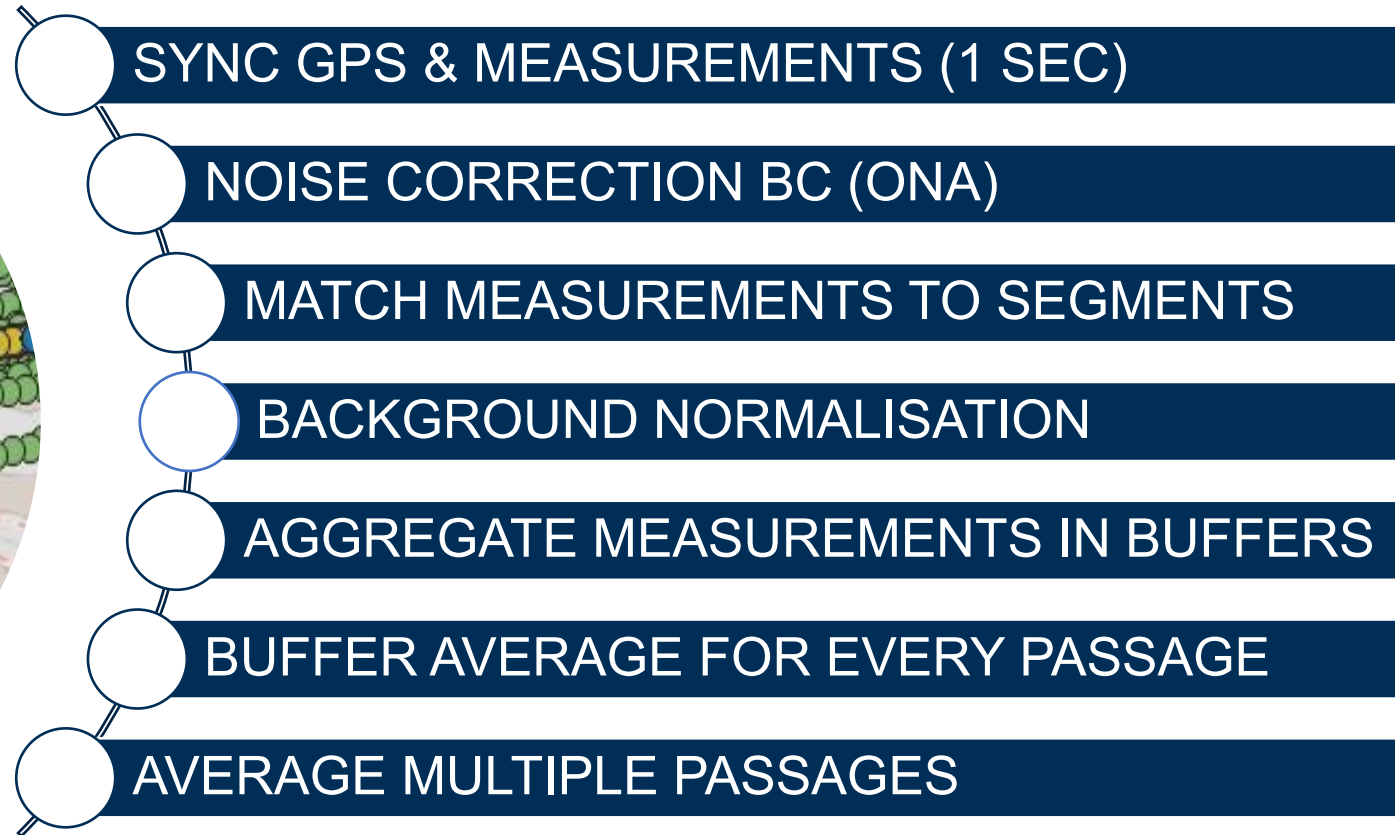
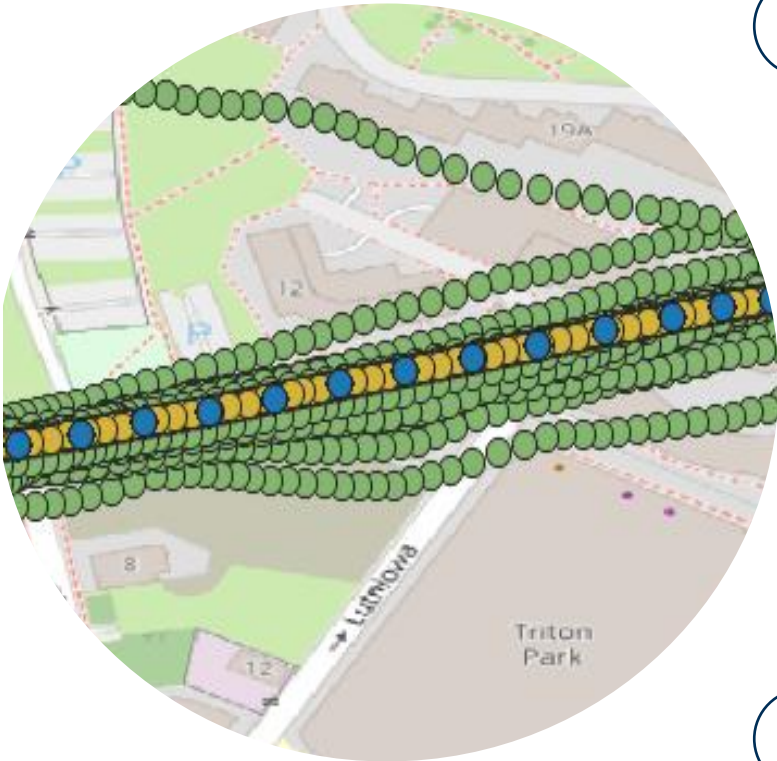
Data processing

- Quality control for sensors
 - Outlier detection (related to e.g. RH)
 - Co-location with reference (representative conditions)
 - Validation instrument
 - If needed: calibration model/ adjustments
- Mobile monitoring
 - Measurements at different locations & times
 - Rescaling is needed to:
 - 1) Extrapolate measured averages to long-term (annual) exposure
 - 2) Rescale data collected at another period in time (~opportunistic)
 - GPS accuracy & time synchronization



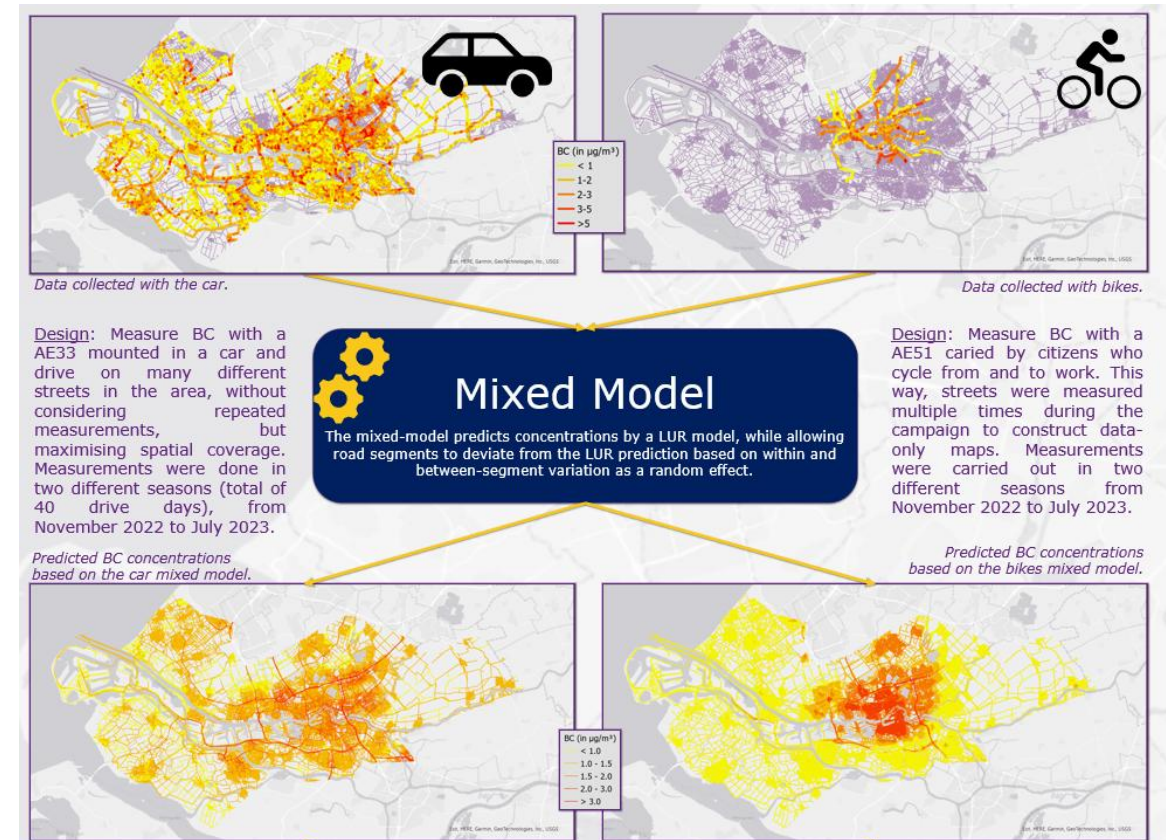
$$BC_{norm,i} = BC_i - BC_{bg,i} + \overline{BC}_{bg}$$

Data processing mobile measurements



Data processing of mobile measurements

- **Data-only vs modelling approach**
- Data-only approach:
 - Mobile data as validation for deterministic models
 - Limited spatial coverage
 - High accuracy/measured
- Modelling approach:
 - Mobile data as input for empirical models
 - Wider spatial coverage
 - Model assumptions



Involvement of citizens

- Instructions!
 - Operation of instruments
 - when and where to collect data
 - Central point of contact
 - Aim of the research
- Engagement and interactions with citizens
 - Building trust, anonymity (ethics)
 - Feedback of results
 - Reward, interest in own environment
- Recruitment process
- Added value of including citizens:
 - Valuable knowledge → domain experts
 - Awareness raising, identification hotspots, representative exposure assessments

Examples of mobile campaigns

Case study Rotterdam

- BC measurements
 - AE51 and GPS
- Commuters of DCMR
 - Opportunistic (but restricted area)
 - During peak hours (mainly)
 - Different road segments collected at different days
- Summer and winter campaign



RESULTS (aggregated data)



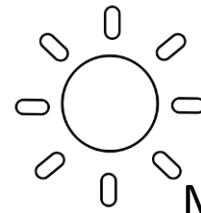
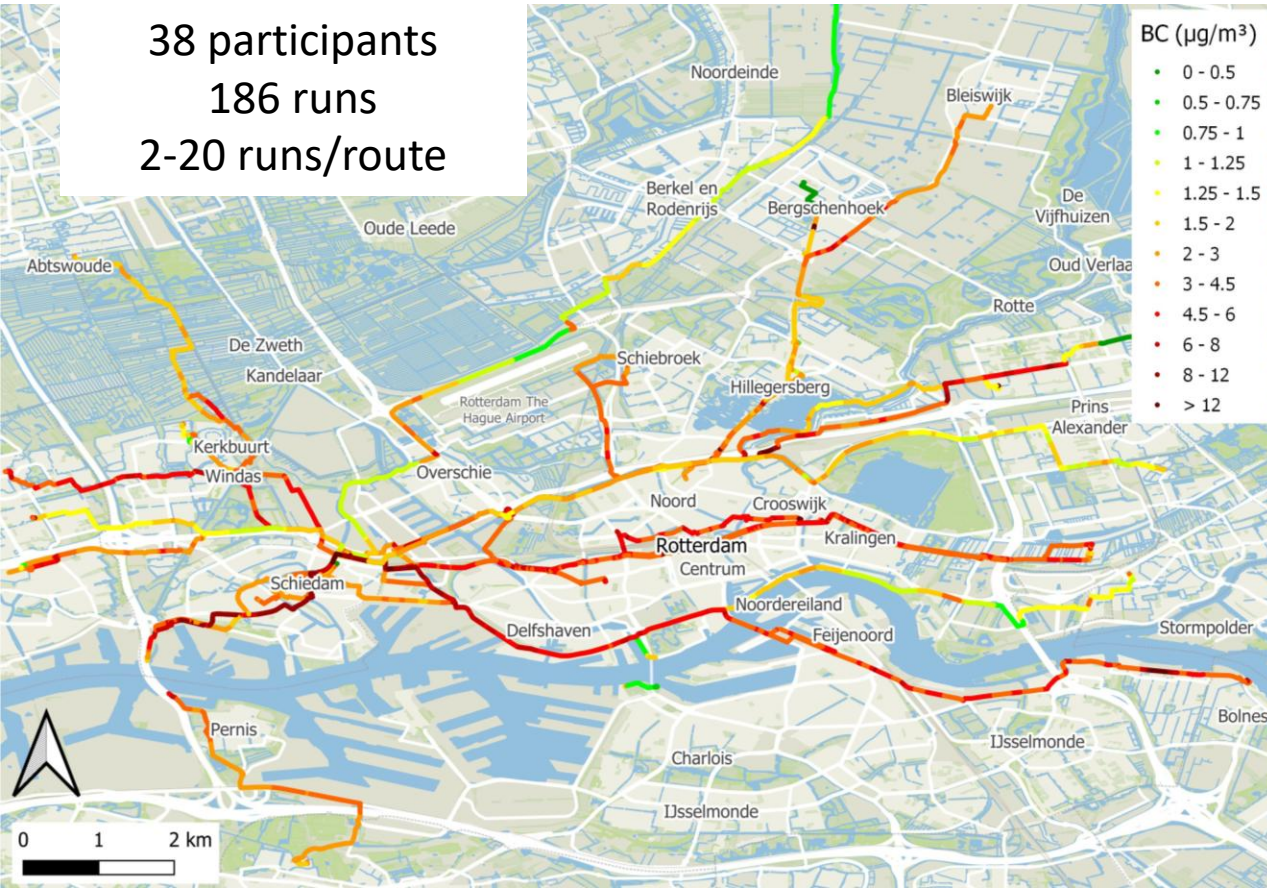
WINTER

Nov '22 – Mar '23

38 participants

186 runs

2-20 runs/route



SUMMER

May '23 – July '23

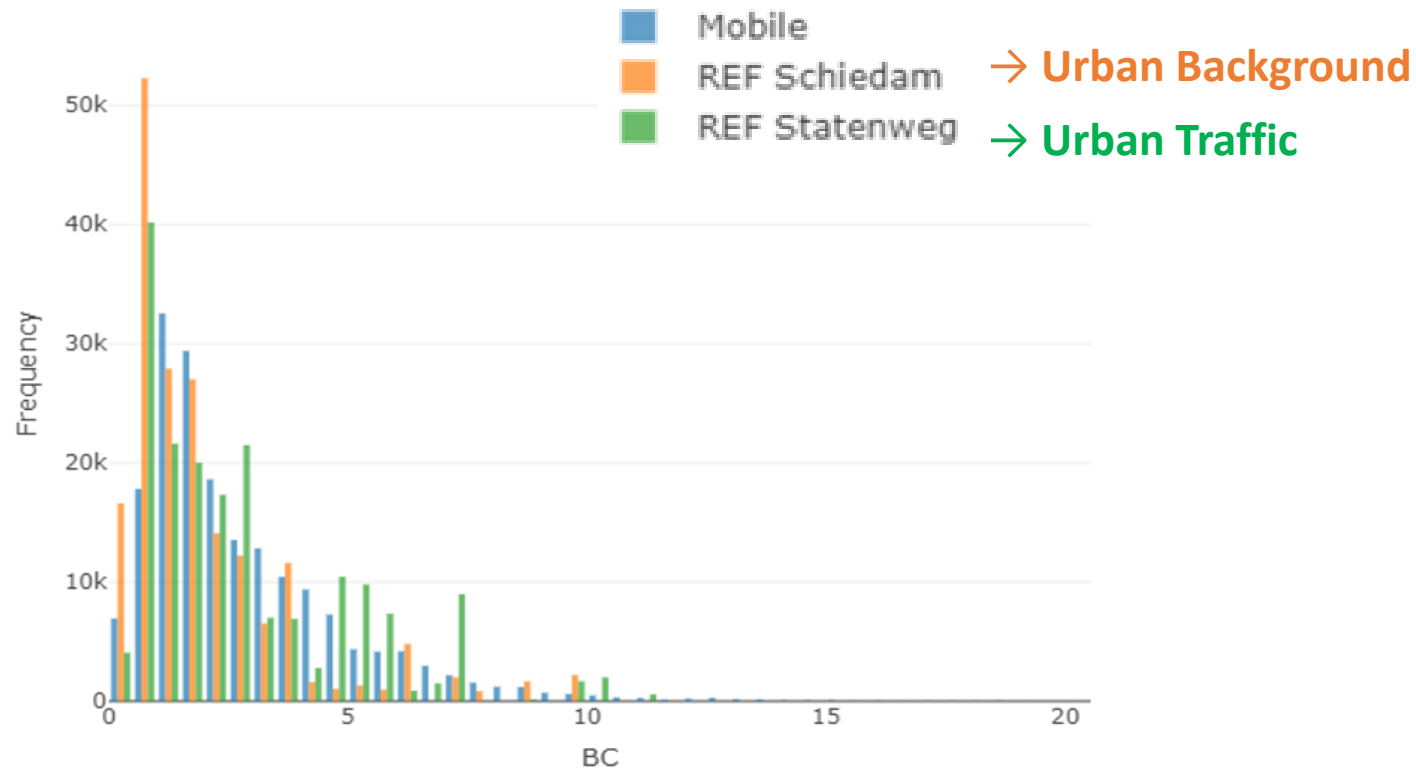
38 participants

186 runs

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Comparison to AQMS

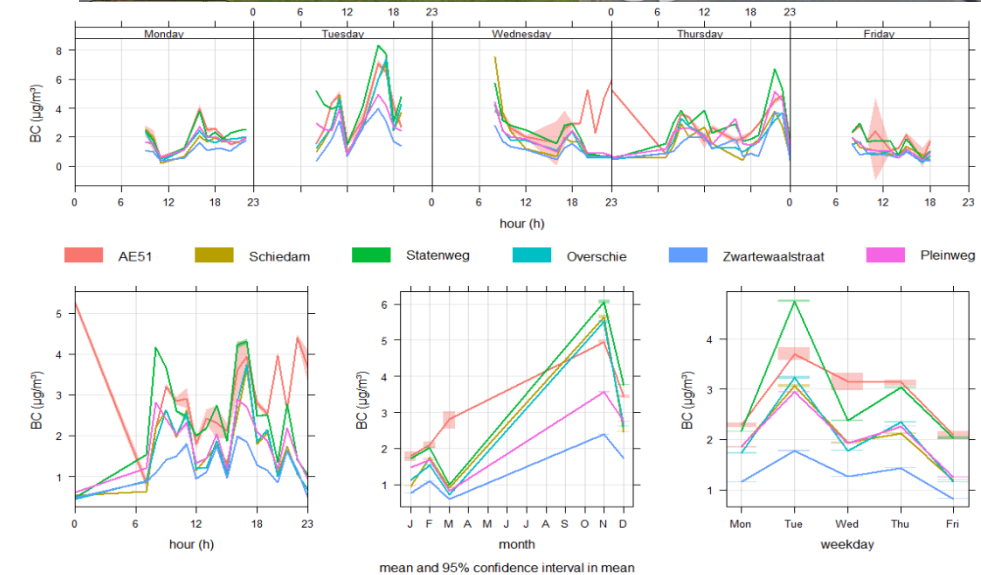


Rescaling with fixed AQMS

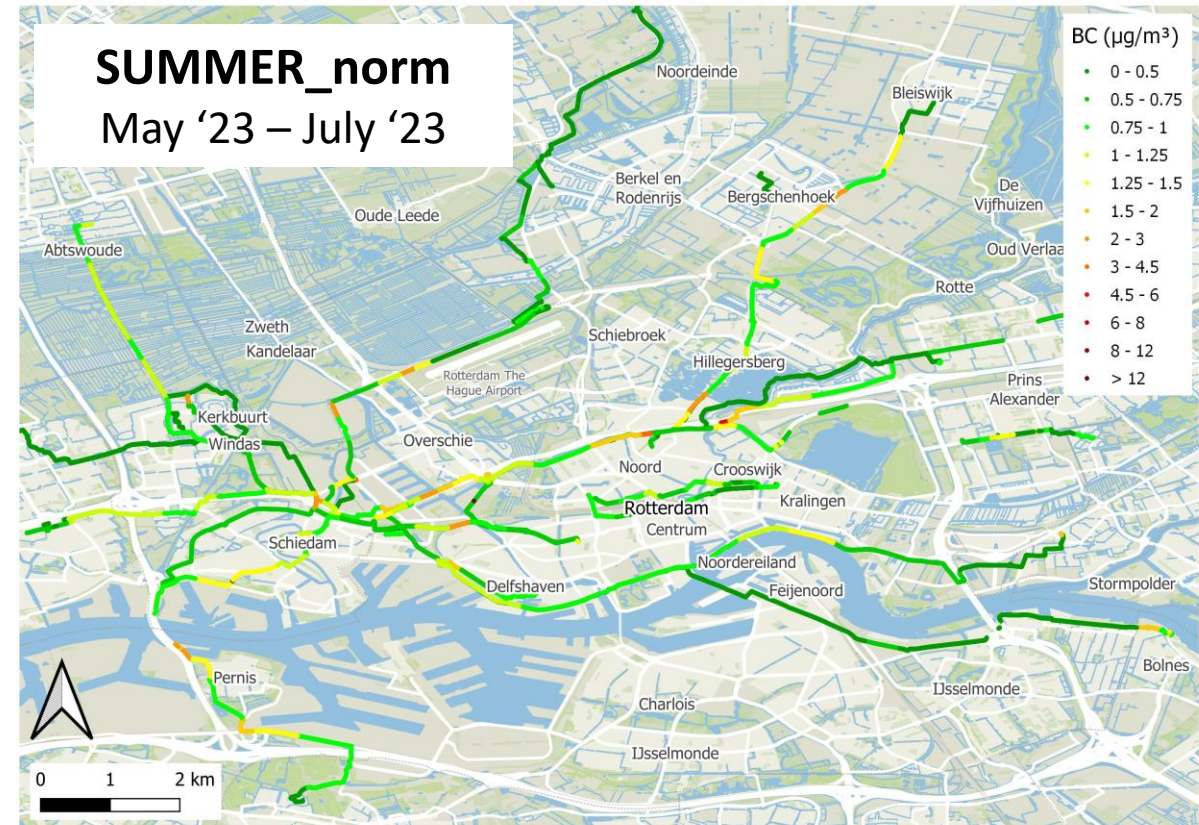
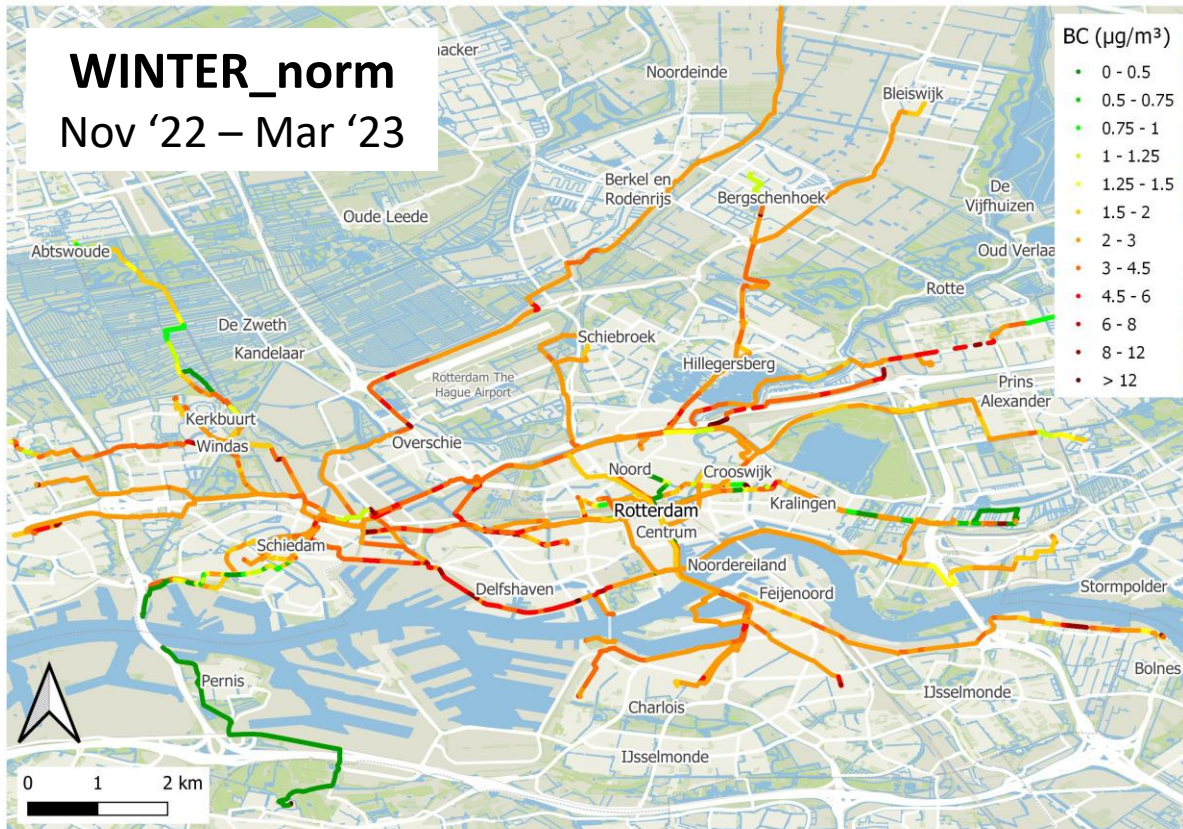
- Comparison of 5 nearby AQMS (REF stations)
- Hourly normalization:
 - Absolute vs **background station** (Schiedam)
 - Relative vs **roadside station** (Statenweg)

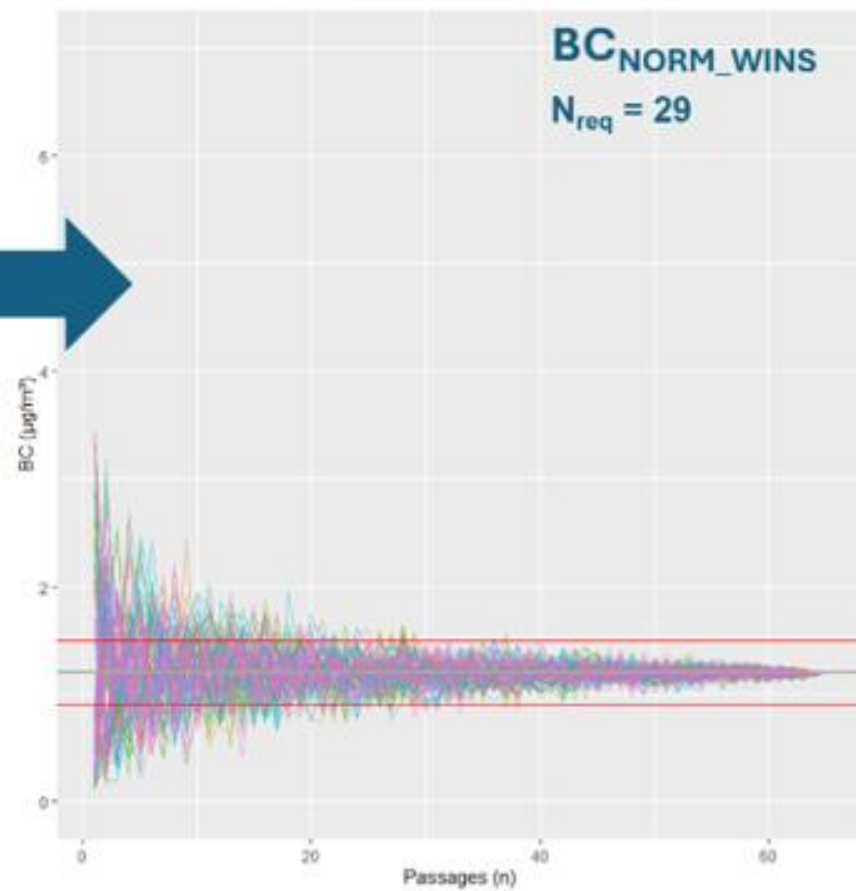
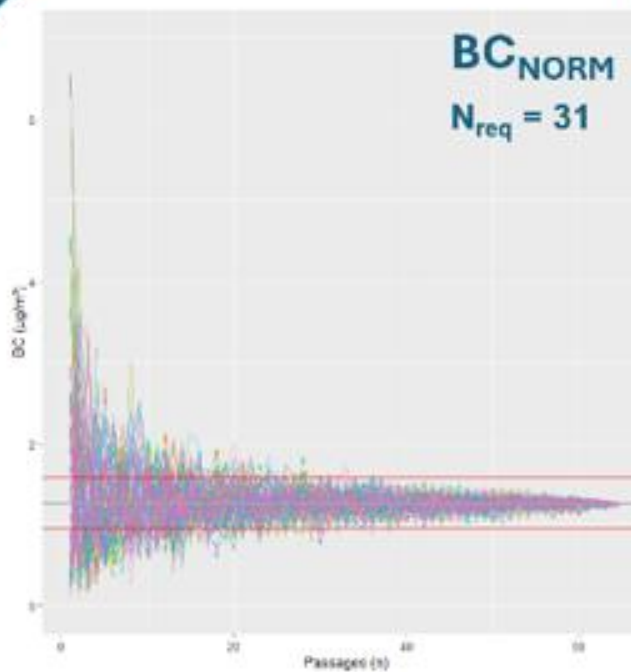
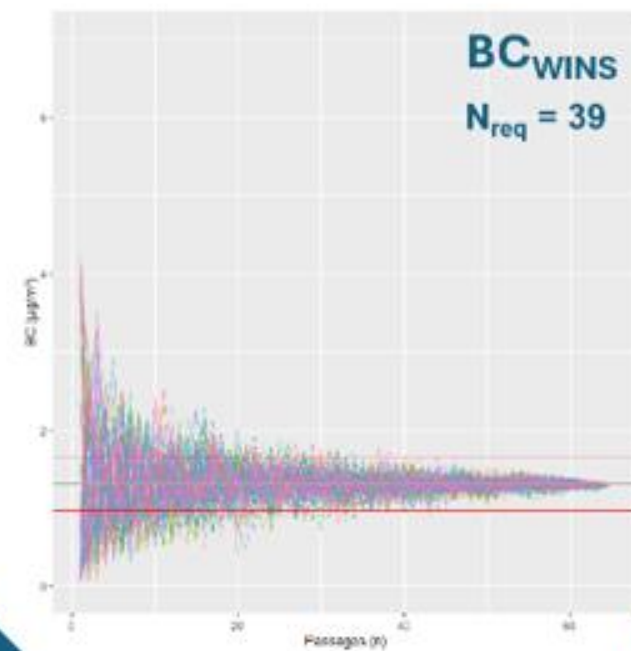
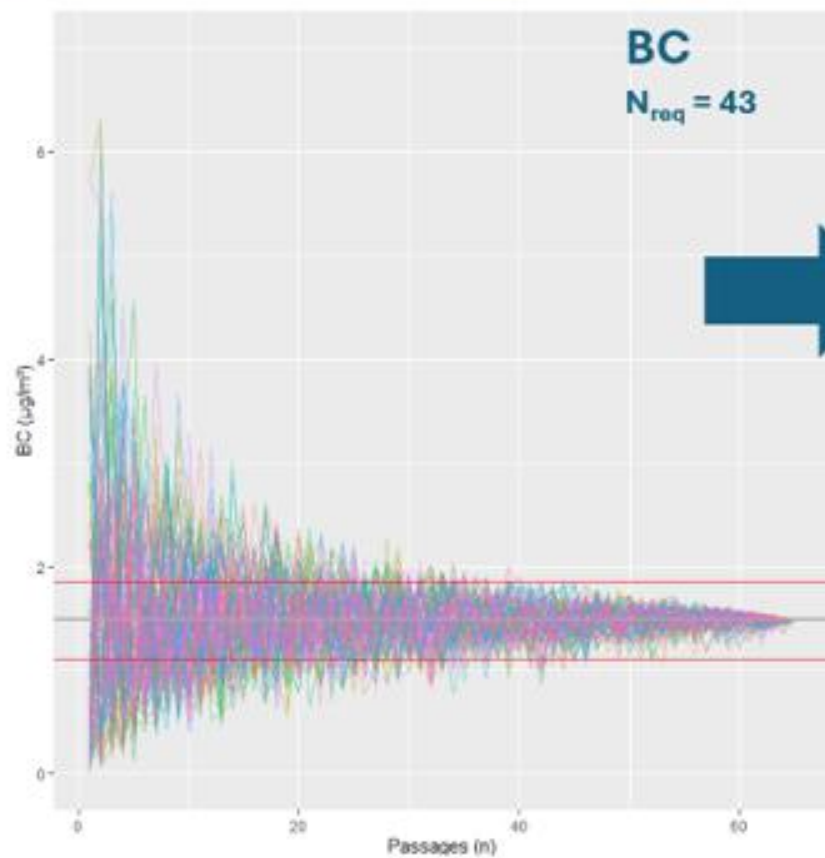
$$BC_{norm,i} = BC_i - BC_{bg,i} + \overline{BC_{bg}} \quad (\text{additive}) \quad (2a)$$

$$BC_{norm,i} = BC_i / BC_{bg,i} \cdot \overline{BC_{bg}} \quad (\text{multiplicative}) \quad (2b)$$

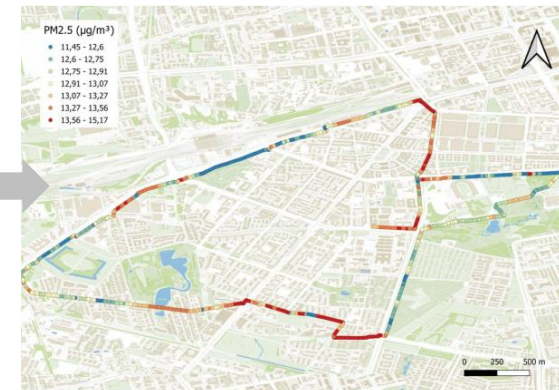
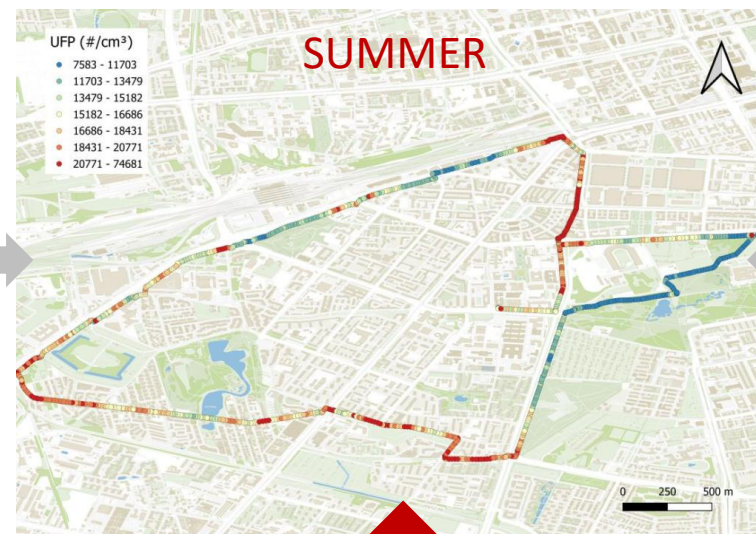
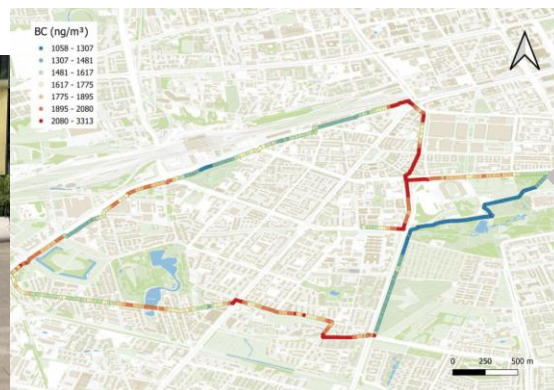


RESULTS (aggregated rescaled data)





Mobile measurements in Warsaw



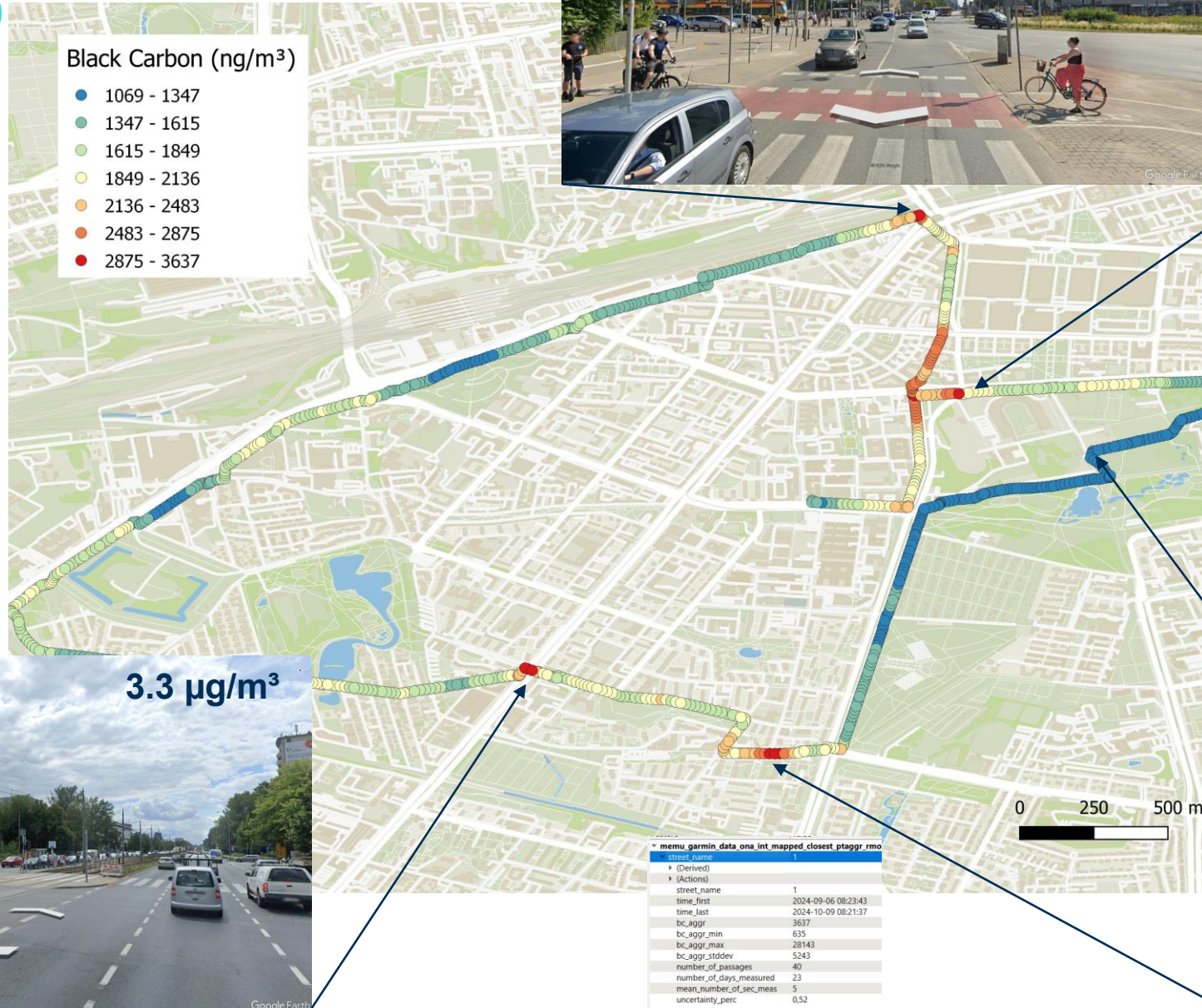
- BC, UFP, PM_{2,5}
- 2 seasons
- Rush hour
- Dedicated (fixed route)
- 40 repeats



Hofman J. et al. in preparation

BC HOT/COLDSPOTS

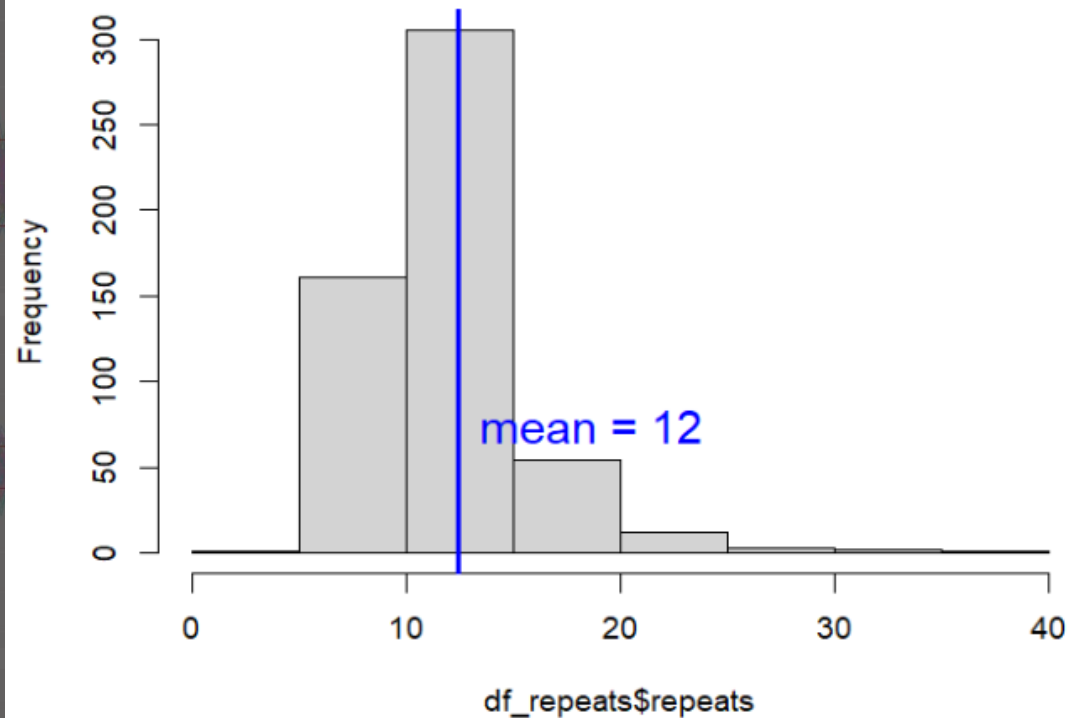
AVG n=40



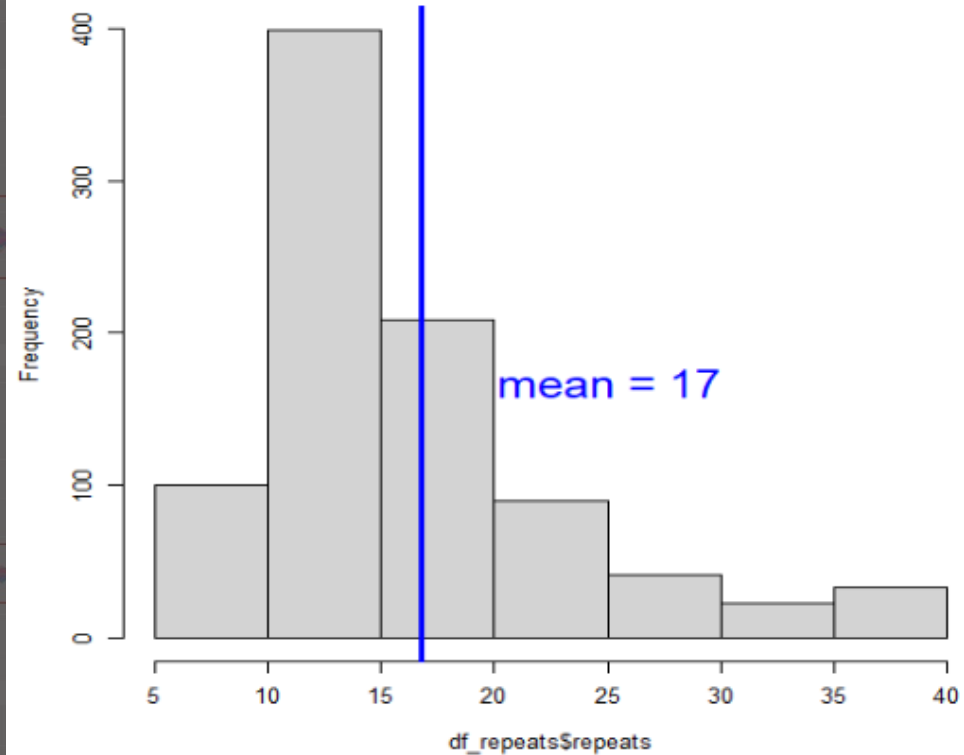
WINTER (n=539)

SUMMER (n=894)

Histogram of df_repeats\$repeats



Histogram of df_repeats\$repeats



Mobile monitoring campaign in Mechelen

- As part of two Citizen science projects:
 - GroundTruth 2.0 and Flamenco
- Aim of campaign:
 - AQ along cycling lanes
 - Policy makers and scientists
- Results:
 - Spatial and temporal variability
 - different per pollutant (PM vs BC)
 - Shows efficiency of traffic restriction measures



Article

Exploring the Spatial Variability of Air Pollution Using Mobile BC Measurements in a Citizen Science Project: A Case Study in Mechelen

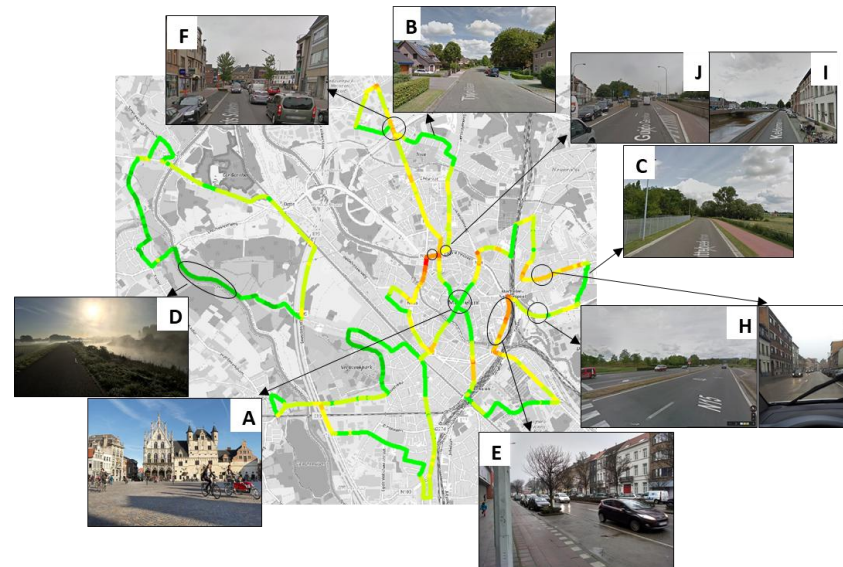
Martine Van Poppel ^{1,*}, Jan Peters ¹, Stijn Vranckx ¹, Jo Van Laer ¹, Jelle Hofman ¹, Bram Vandeninden ^{2,3}, Charlotte Vanpoucke ² and Wouter Lefebvre ¹

¹ Flemish Institute for Technological Research (VITO), Boeretang 200, 2400 Mol, Belgium

² Belgian Interregional Environment Agency, Gaucheretstraat 92-94, 1030 Brussels, Belgium

³ School of Public Health, Université Libre de Bruxelles, Route de Lennik 808-CP591, 1070 Brussels, Belgium

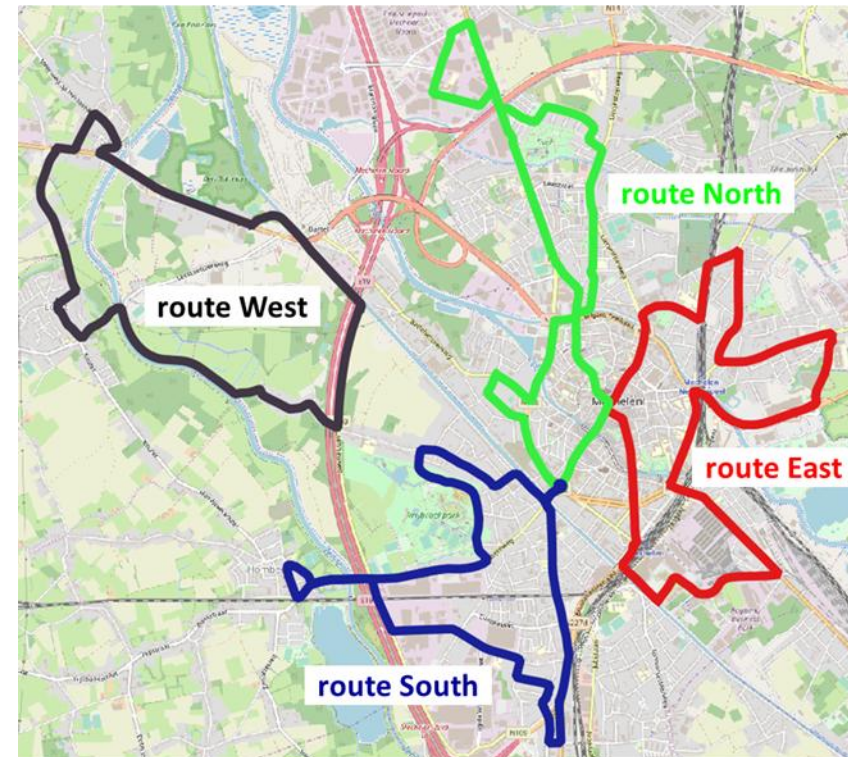
* Correspondence: martine.vanpoppel@vito.be



Measurement approach

- Along 4 routes in Mechelen: 40 km
- During morning and evening rush hour
- In 4 campaigns of 2 weeks (± 25 runs): 280 h
- Simultaneous measurements (except route South in Campaign 1)
- 50 volunteers/2800 km!

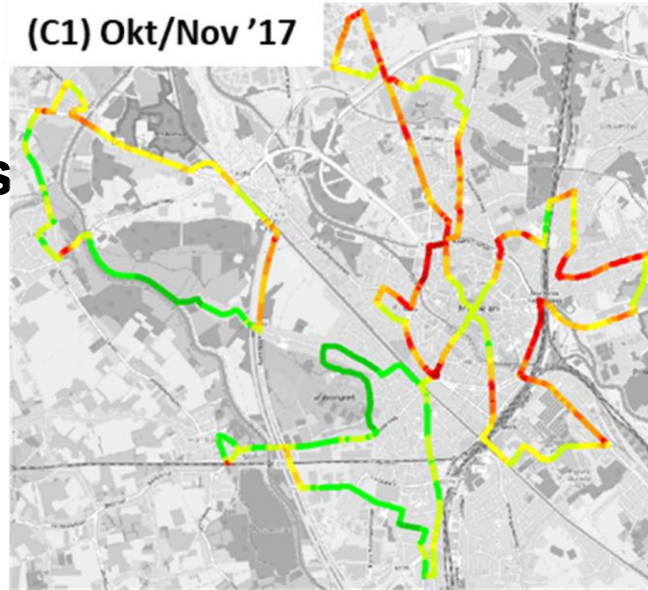
Campaign	Wind direction	Wind speed	Temperature
23/10/17 – 05/11/17	SW	3.5 m/s	10.5 °C
06/11/17 – 19/11/17	NW	2.8 m/s	5.9 °C
21/02/18 – 06/03/18	NE	4.9 m/s	-0.6 C
25/06/18 – 08/07/18	NE	3.9 m/s	22.7 °C
17/09/18 – 30/09/18	SW	3.1 m/s	13.6 °C



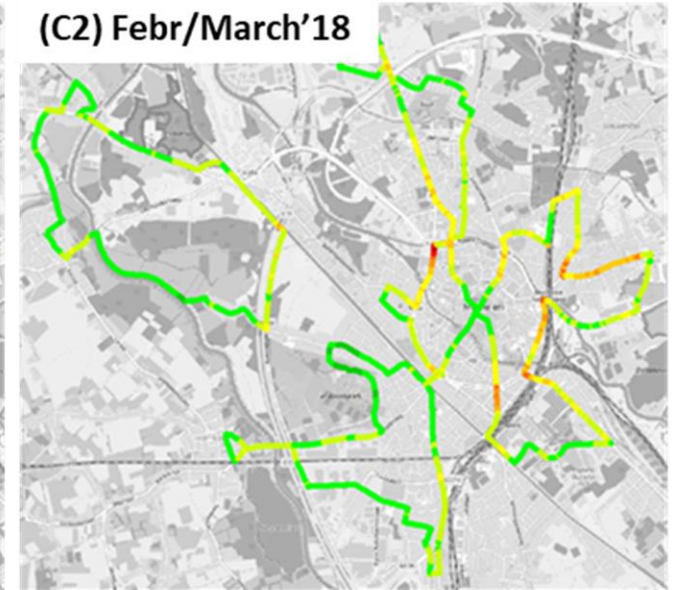
Results (4 campaigns)

- Large differences
 - Between locations
 - Between seasons
 - Similar pattern

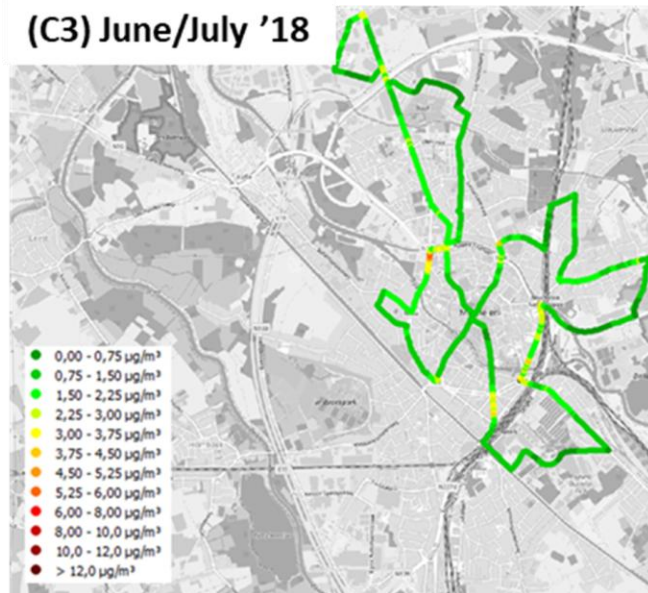
(C1) Okt/Nov '17



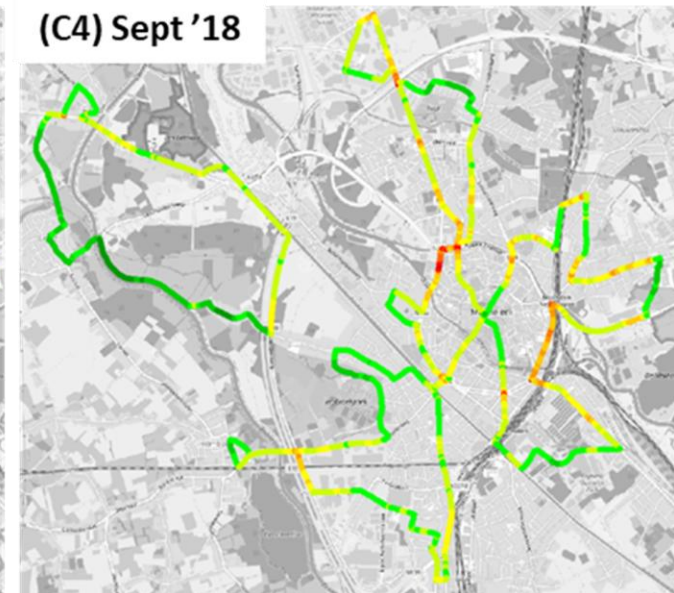
(C2) Febr/March '18



(C3) June/July '18



(C4) Sept '18



Comparison with fixed AQM stations

- Summary statistics per campaign and BC concentrations at fixed monitoring stations (VMM)

	Min.	1 st Qu.	Median	Mean	3 rd Qu.	Max.	VMM virtual Traffic station
Campaign 1	1.0	2.4	3.3	3.7	4.7	12.4	3.3
Campaign 2	1.3	2.0	2.3	2.5	2.9	10.5	2.4
Campaign 3	0.5	0.9	1.2	1.4	1.7	6.2	1.4
Campaign 4	0.5	1.8	2.5	2.5	3.1	8.3	2.5
				Yearly average 2017: 1.9			

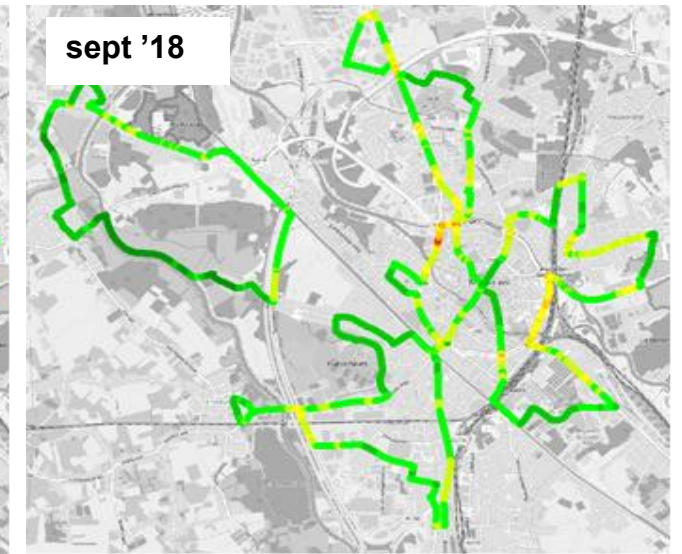
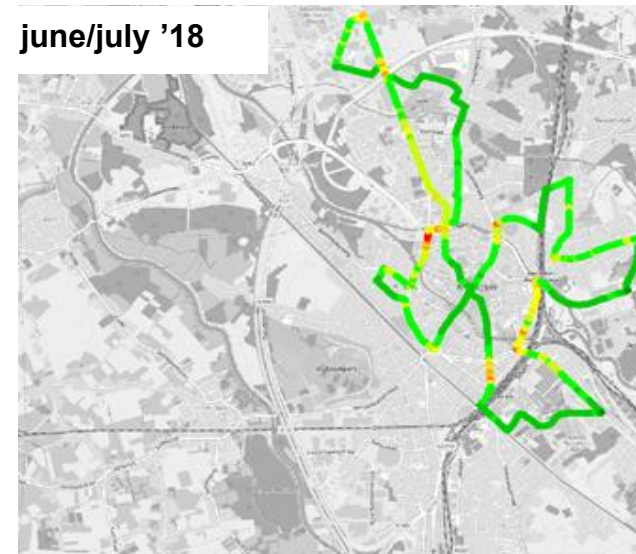
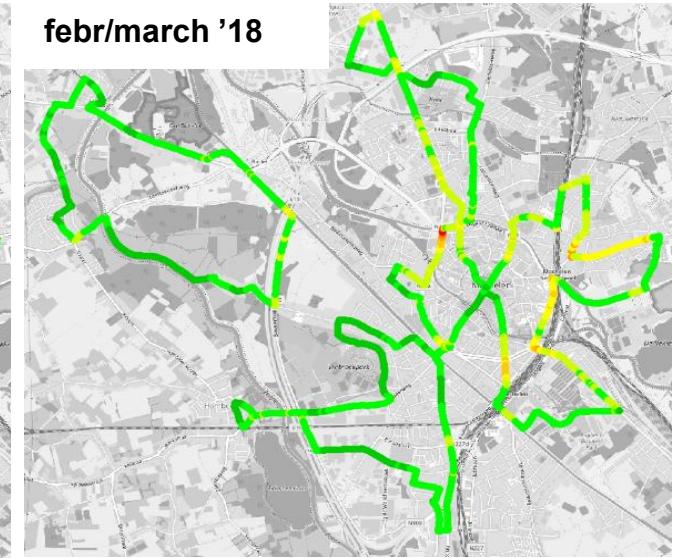
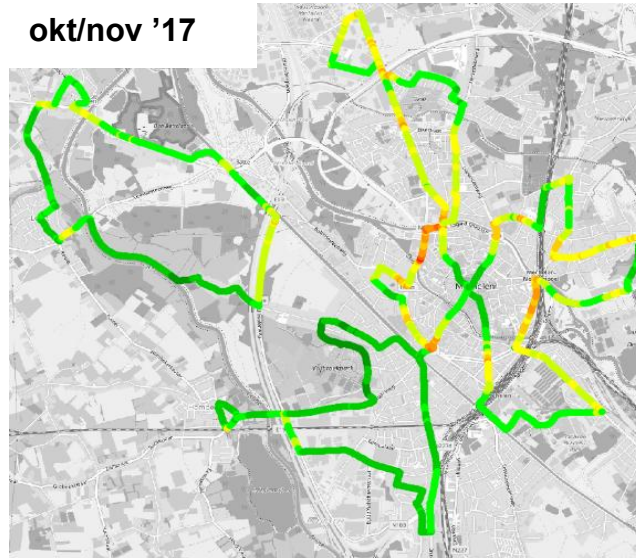
→ Max concentration

→ Min concentration

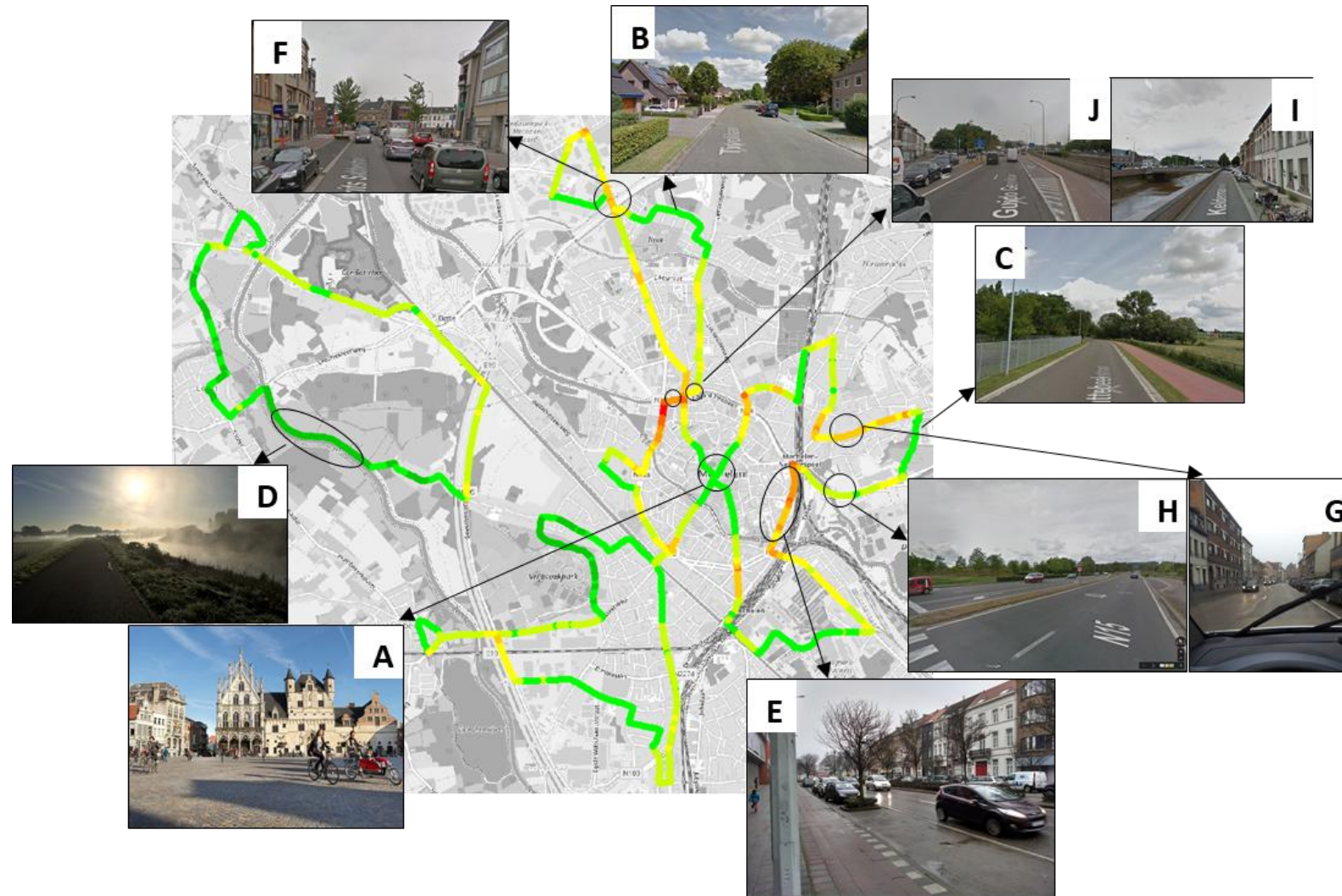
- Rescaling** the map based on data on 'virtual' traffic station of the air quality monitoring network :
ratio = yearly average/campaign average @ VMM traffic station
- Aggregation** of different campaigns based on weighed average (weighing factor ~ frequency of occurrence of the average BC concentration)

Results (rescaled)

- Factors yearly av/campaign av:
0,57 – 1,37
- More similar pattern
- Holiday season is less suitable to
assess impact of traffic
- Cold season can be impacted by
other sources



Results (aggregated map)



Mobile mapping to assess impact of traffic measures

- Campaign in Cluj-Napoca (Romania)
- Spatial variability: Increased BC concentrations at traffic locations
- Reduced concentrations during traffic restriction (COVID pandemic)
- Background corrections needed to assess impact



Mobile measurements of black carbon: Comparison of normal traffic with reduced traffic conditions during COVID-19 lock-down

Martine Van Poppel^{a,*}, Jan Peters^a, Erika Andrea Levei^b, Luminița Mărmureanu^c, Ana Moldovan^b, Maria-Alexandra Hoaghia^b, Cerasel Varaticeanu^b, Jo Van Laer^a

^a Flemish Institute for Technological Research (VITO), Boeretang 200, 2400, Mol, Belgium

^b Research Institute for Analytical Instrumentation Subsidiary, National Institute of Research and Development for Optoelectronics INOE 2000, 67 Donath, RO400293, Cluj-Napoca, Romania

^c Remote Sensing Department, National Institute of Research and Development for Optoelectronics INOE2000, 409 Atomigilor, RO077125, Măgurele, Ilfov, Romania

HIGHLIGHTS

- Black Carbon (BC) concentration in Cluj-Napoca depend on the traffic intensity.
- Traffic restriction due to COVID-19 pandemic reduced BC concentrations.
- Mobile measurements can be used to assess spatial variability of BC.
- Background corrections are needed to assess the impact of traffic measures on BC.
- Traffic-free zones along the routes can be used for background corrections of BC.

GRAPHICAL ABSTRACT



Some take-away messages

- Different approaches can be used and are function of use case
- Best practices:
 - Validation instruments
 - Co-location campaign
 - between instruments (BSU) + comparison against REF
 - ideally on-site/similar conditions
 - Data collection:
 - Dedicated vs opportunistic
 - Number of required repeats (e.g. >15 within season vs >30 multi-season)
 - Data analysis
 - Time/GPS synchronization
 - Data cleaning
 - Post-processing (Map matching/background normalization/winsorizing/...)
 - Number of required repeats (measurements only vs model approach)
 - With citizens
 - Valuable knowledge → domain experts
 - Single POC/communication is key/templates/on-site training
 - Provides a lot data!

